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CAMSCKF: A Multi-State Constraint Kalman Filter with Adaptive Multivariate Noise Parameters Clustering and Estimation for Visual-Inertial Odometry

Yiyang Tang, Hanxuan Zhang, Yichen Yu, Xiaofeng Li and Yulong Huang

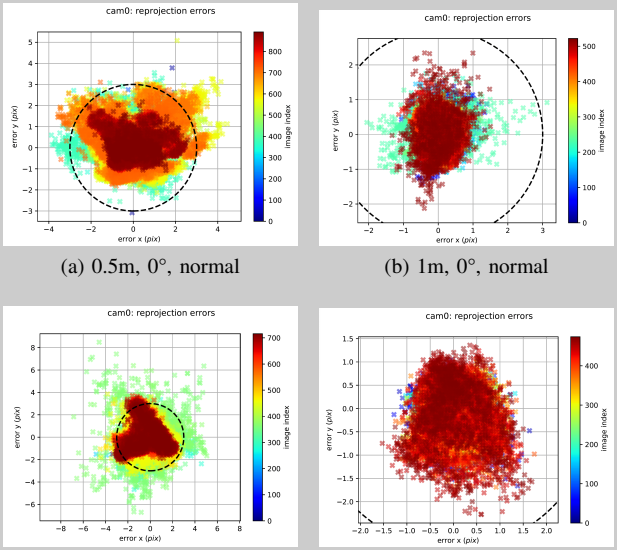
***Abstract*— The Visual-Inertial Odometry has been widely deployed on autonomous robots traveling in open outdoor scenarios. However, the visual measurements will be influenced heavily by the observation distances, perspectives, lighting and texture conditions, with distinct and time-varying noise distributions of measurements. Existing methods for handling time-varying noise in Visual-Inertial Odometry regard all measurement noise as identically distributed, unable to effectively deal with the distinct noise in open outdoor scenarios, which degrades the localization accuracy. In this paper, a Multi-State Constraint Kalman Filter with Adaptive multivariate noise parameters Clustering and estimation for visual-inertial odometry (CAMSCKF) is proposed to address the issue, which can separately track the measurement noise covariance matrix (MNCM) of different measurement clusters and adjust the MNCM in real-time. Firstly, the joint distribution of the state and the MNCM coefficients for each cluster is modeled as an Gaussian-Multivariate Generalized Inverse Gaussian distribution. Subsequently, an Expectation Maximization algorithm-based stepwise adaptive measurement clustering method is designed, which clusters measurements according to their corresponding innovations. Finally, an analytical update method for the joint posterior distribution without fixed-point iteration is implemented, achieving adaptive adjustment of the MNCM, thereby enabling accurate and robust Visual-Inertial Odometry localization. The superiority of the proposed method is demonstrated by simulations and dataset experiments, especially under the aggressive motion. In the experiments of the challenging outdoor dataset UZH-FPV, the proposed method has improved the average position and attitude estimation accuracy by 35.69% and 32.88%, respectively, compared with the state-of-the-art ANGIG-KF.**

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I. INTRODUCTION

A. Background

Accurate and robust localization is crucial for autonomous robots traveling in open outdoor scenarios. The Visual-Inertial Odometry (VIO) has become one of the mainstream solutions for autonomous robots for its capability to achieve accurate and robust localization without relying on the external global information. There have been numerous mature VIO methods, which can be mainly divided into two categories according to their back-ends, i. e., optimization-based [1]-[5] and filter-based VIO [6]-[9]. Although the optimization-based VIO holds the merits of satisfactory scalability and higher localization accuracy, it relies on sufficient computational resources heavily. In contrast, the filter-based VIO has the advantages of high computational efficiency and low memory footprint, which is suitable for autonomous robots with resource limitations [8]. Among various filter-based VIO methods, MSCKF [6] has been widely applied in robot localization due to its high computational efficiency and satisfactory accuracy. These advantages profit from the unique update strategy, which overcomes the high computational cost and dimension explosion problems of the traditional EKF-SLAM [10].



(a) 0.5m, 0°, normal (b) 1m, 0°, normal

(c) 0.5m, 0°, weak (d) 0.5m, 60°, normal

Fig. 1: A diagram illustrating the variation of feature point noise distribution in different conditions (observation distance, perspectives and lighting condition). The radius of the dashed circle is 3 pixels, which is demonstrated for reference.

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However, in open outdoor environments with dynamic disturbances, the noise distribution of visual feature points in VIO is not only time-varying but also varies across different measurement clusters. For instance, different observation distances, perspectives, lighting and texture conditions can lead to distinct and time-varying noise distributions among different types of feature points [11]. As illustrated in Fig. 1, the covariance of measurement noise is influenced by different observation distances, perspectives and lighting conditions. It leads to a significant mismatch between the

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actual and nominal MNCM, further introducing the measurement errors to the estimation of state through MNCM, thus declining the localization accuracy and robustness of VIO. Although some robust methods commonly used in the front-end, such as RANSAC and Mahalanobis gating, can achieve normalization of the measurement noise covariance matrix to some extent by filtering outliers, they are sensitive to thresholds and fail to solve the problem that MNCM varies under different observation conditions. To address the aforementioned issues, This paper deals with the problems that the front-end cannot fully solve in the back-end, and proposes ACMSCKF to cluster feature points based on measurement noise and track the noise statistical characteristics of each cluster with different Gaussian distribution. By utilizing a analytical joint distribution update method, the joint estimation of the state and multi-cluster noise parameters is realized, achieving accurate and robust localization for VIO.

B. Related Work

In recent decades, numerous well-developed VIO methods have been proposed [1]-[10]. Among these methods, MSCKF [6] presents an extended Kalman filter-basd tightly-coupled VIO framework, which addresses the dimension explosion problem of state vector in EKF-SLAM [10]. OKVIS [12] fuses the preintegration residual of the IMU and the reprojection residual during the optimization process, achieving satisfactory state estimation. ORB-SLAM3 [13] is presented as a SLAM system for monocular, stereo and RGB-D cameras to achieve high-accuracy localization. VINS-Mono [14] integrates IMU preintegrated measurements and feature observations via a tightly coupled, nonlinear optimization method, which enables accurate VIO localization. Although the above mentioned visual-inertial methods work well in normal environments, they suffer from significant decline in accuracy and robustness once confronted with dynamic disturbances in open outdoor environments because of the mismatch between nominal values and true values of MNCM [15][16].

Adaptive filters were developed to deal with the time-varying problem of noise parameters, which can be mainly divided into three types, i. e., covariance matching method [17][18], maximum likelihood method [19][20] and Variational Bayesian (VB) method [21]-[24]. Unfortunately, covariance matching methods and maximum likelihood methods have been proven to be unable to guarantee convergence and realize the estimation of time-varying noise parameters. Adaptive filter based on VB become the mainstream for its better robustness and accuracy. The VB-MSCKF [24] models the MNCM as an Inverse-Wishart distribution, enabling adaptive adjustment of the MNCM. However, the Frobenius norm of the matrix is employed in VB-MSCKF for approximate calculations, which leads to large approximation errors. ANGIG-KF [25] solves this problem by modeling the MNCM as a product of the scalar coefficient and the identity matrix, the MNCM coefficients are modeled as Generalized Inverse Gaussian (GIG) distributions, thus achieving joint estimation of the MNCM coefficients and the state under the framework of VB. Although ANGIG-KF can track the variation of environmental noise distribution to a certain extent, its strict assumption that all feature points hold the same noise distribution makes it unable to handle distinct noise distribution variations, which decreases the accuracy and causes instability in outdoor dynamic scenarios.

Some robust filters have been developed to address the challenge of disturbance on measurement. The Huber kernel-based [26] and the Student's t distribution-based [27] robust filters have been proven to be effective. These methods can assign weights to measurements dynamically, reducing the influence of disturbances. A robust interactive multiple model-based VIO is proposed to deal with uneven lighting, feature loss and texture duplication situations [11]. In [28], a H∞-based VIO has also been proposed, achieving stable localization performance even if the measurement noise distribution is unknown. However, these robust filter-based VIO can't continuously track the noise variances of the measurements, resulting in redundant iterations, which limits the computational efficiency and localization accuracy.

C. Motivations

It is obvious that the performances of the existing adaptive-based and robust-based VIO are limited when facing distinct and time-varying measurement noise which exists in open outdoor scenarios. Some methods like the state-of-the-art ANGIG-KF, perform well in normal scenarios but suffer from the following drawbacks in open outdoor scenarios: (a) Existing adaptive filter-based VIO methods assume that all measurements hold the same noise variance distribution. However, in open outdoor environments, the measurement noise of different feature point clusters is not identically distributed. Estimation with the noise covariance modeled by the ideal modeling will lead to a mismatch between the estimated MNCM and the actual one, resulting in a decrease in localization accuracy. (b) Although existing robust filter-based VIO methods can process measurements with different noise variances and adjust weights dynamically, the noise variances of the measurements cannot be tracked continuously. Only current measurements can be processed, and the historical information is discarded, resulting in unstable localization performance. To address the problem above, a general adaptive VIO framework is proposed, which aims to handle and track the measurements with different noise variances continuously.

D. Contributions

In this paper, the ACMSCKF is proposed, which can handle and track the measurements with different noise variances continuously. Firstly, the joint distribution of the state and the MNCM coefficients for each cluster is modeled as the Gaussian-Multivariate Generalized Inverse Gaussian (GMGIG) distribution. Then, an Expectation Maximization (EM) algorithm-based stepwise adaptive measurement clustering method is designed, which employs a chi-square test for pre-clustering measurements to reduce computational

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cost, enabling the autonomous generation of clusters. Subsequently, the indicator function for the cluster membership of each measurement is introduced, and the online cluster of measurements is achieved by employing the EM algorithm. Finally, an analytical update method for the joint posterior distribution without fixed-point iteration is carried out. It enables the tracking of the measurement noise distribution of each cluster and achieves automatic adjustment of MNCM. The main contributions are listed below

- To address the challenge of non-identically distributed measurement noise of feature points for MNCM estimation, an EM algorithm-based stepwise adaptive measurement clustering method is designed. The measurements can be clustered online based on their innovations, thus facilitating the estimation of the MNCM.
- To accurately track the measurement noise covariance of each cluster, the joint distribution of the MNCM coefficients and state is modeled as an GMGIG distribution. An analytical update method for the joint posterior distribution without fixed-point iteration is implemented, enabling the accurate tracking of the measurement noise covariance and the adaptive adjustment of the MNCM, thereby realizing accurate and robust localization in open outdoor scenarios with time-varying noise.
- A sufficient number of simulations and dataset experiments have been conducted, which verifies that the proposed method can significantly improve the localization accuracy and robustness of VIO with time-varying measurement noises.

The structure of this paper is as follows. The fundamental principle and workflow of the proposed ACMSCKF are given in Section II. Simulations and dataset experiments are carried out in Sections III and IV, respectively. This paper is concluded in Section V.

II. PROPOSED ACMSCKF

In this section, the fundamental principle and workflow of the ACMSCKF are introduced. Firstly, the joint probability density function (PDF) of the state and the MNCM coefficients is modeled as GMGIG distribution. Subsequently, an EM algorithm-based stepwise adaptive measurement clustering method is introduced, which clusters measurements according to their corresponding innovations. Finally, an analytical update method for the joint posterior distribution without fixed-point iteration is implemented, achieving adaptive adjustment of the MNCM, thereby enabling accurate and robust VIO localization. The implementation of the proposed ACMSCKF is presented in Fig. 2.

A. Linear State-space Model with Multi-Cluster Measurement Noises

In this subsection, a linear state-space model with multi-cluster measurement noises is established. The state-space model is formulated as follows

$$\begin{cases} \boldsymbol{x}_k = \boldsymbol{\Phi}_k \boldsymbol{x}_{k-1} + \boldsymbol{G}_k \boldsymbol{\omega}_k \\ \boldsymbol{z}_k = \boldsymbol{H}_k \boldsymbol{x}_k + \boldsymbol{v}_k \end{cases} \tag{1}$$

where $\boldsymbol{x}_k$ and $\boldsymbol{z}_k$ denotes the state vector and the measurement vector at time step k, $\boldsymbol{\Phi}_k$, $\boldsymbol{G}_k$ and $\boldsymbol{H}_k$ are the state transition matrix, process noise coefficient matrix and observation matrix, respectively. The definition of $\boldsymbol{x}_k$ and the forms of $\boldsymbol{\Phi}_k$, $\boldsymbol{G}_k$ and $\boldsymbol{H}_k$ are consistent with Tra-MSCKF [6], where $\boldsymbol{x}_k$ contains the estimation errors of attitude, velocity and position, the biases of gyroscope and accelerometer, and the clone states in the sliding window. The process noise $\boldsymbol{\omega}_k$ is characterized as zero-mean Gaussian noise, i.e., $\boldsymbol{\omega}_k \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}_k)$.

Moreover, as mentioned earlier, the actual visual measurements are observed under different distances, perspectives and lighting conditions, which actually result in different variances of each dimension of the measurement noise vector. Different from other methods that assume the measurement noise vector follows a multivariate Gaussian distribution with the same variances across all dimensions, the measurement noise vector is modeled as a multivariate Gaussian distribution innovatively in this paper, where the variances of each dimension are not exactly the same. As shown in Fig. 3, each dimension of the measurement noise vector will be divided into different clusters, following a Gaussian distribution with zero-mean and different variances. The PDF of the measurement noise is modeled as follows

$$\begin{cases} \boldsymbol{v}_k \sim \mathcal{N}(\boldsymbol{v}_k; \boldsymbol{0}, \boldsymbol{R}_k) \\ \boldsymbol{R}_k = \boldsymbol{Diag}\{\boldsymbol{R}_k^1, ..., \boldsymbol{R}_k^n, ..., \boldsymbol{R}_k^N\} \\ \boldsymbol{R}_k^n = \lambda_k^n \boldsymbol{I}_k^n \end{cases} \tag{2}$$

where λ_k^n denotes the time-varying MNCM coefficient, $\boldsymbol{I}_k^n$ is the identity matrix related to the $n\ th$ cluster, with the same dimension as $\boldsymbol{R}_k^n$.

B. Gaussian-multivariate GIG Distribution-based Model

Since the distribution of the MNCM coefficients needs to be predetermined, the joint distribution model of the state and the MNCM coefficients is established firstly. As a generalized distribution family, the GIG distribution includes the inverse Gamma (IG), Gamma, exponential and numerous other distributions as its special cases. The IG or Gamma distribution is only a special case when the parameters of the GIG distribution meet a certain relationship. Therefore, the parameters of the GIG distribution are more flexible. Thus, the GIG distribution is selected for the MNCM coefficients. The form of GIG distribution can be found in Appendix A. The joint posterior distribution of the state vector $\boldsymbol{x}_k$ and the clustered MNCM coefficients λ_k at time k is established, obeying GMGIG distribution as follows

$$p(\boldsymbol{x}_k, \lambda_k^{1:N} | \boldsymbol{z}_{1:k}) = \underbrace{\mathcal{N}(\boldsymbol{x}_k; \hat{\boldsymbol{x}}_k(\lambda_k^{1:N}), \hat{\boldsymbol{P}}_k(\lambda_k^{1:N}))}_{p(\boldsymbol{x}_k | \lambda_k^{1:N}, \boldsymbol{z}_{1:k})} \times \underbrace{\prod_{n=1}^{N} GIG(\lambda_k^n; \hat{\alpha}_k^n, \hat{\beta}_k^n, \hat{\gamma}_k^n)}_{p(\lambda_k^{1:N} | \boldsymbol{z}_{1:k})} \tag{3}$$

where α_k^n, β_k^n and γ_k^n denote the parameters of the GIG distribution for $n\ th$ cluster measurement noise, $\hat{\boldsymbol{x}}_k(\lambda_k^{1:N})$

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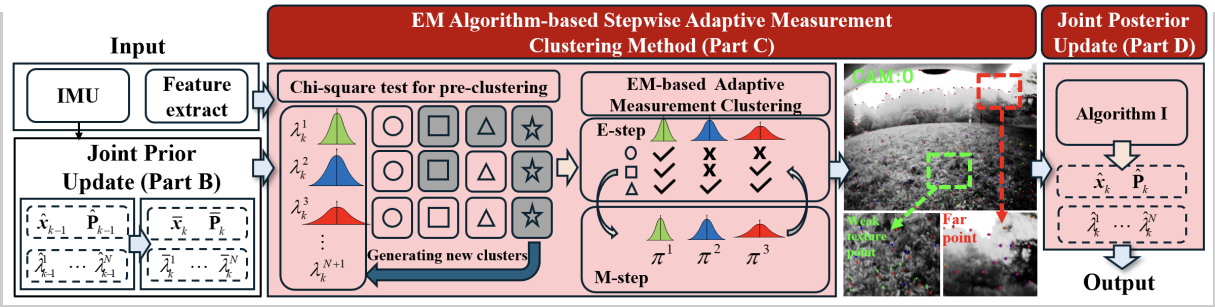


Fig. 2: The implementation of proposed ACMSCKF for VIO, where the red parts are main contributions of this paper. The variables are consistent with the definitions of the corresponding variables in Section II. In Part C, different icons represent different measurements, and the distributions with different colors represent the noise distributions corresponding to different clusters. In the chi-square test pre-clustering, gray block indicates that the measurement which does not pass the chi-test. When there are measurements that not pass the chi-square test for all existing clusters, a new cluster will be generated. The checkmark indicates that the corresponding measurement will be utilized to calculate the probability of belonging to a certain cluster, while the crossmark indicates that the probability will not be calculated.

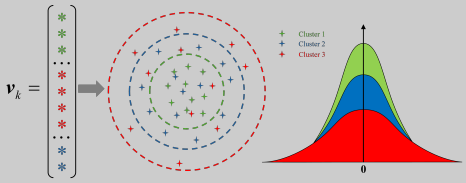


Fig. 3: The diagram that illustrates the components of the measurement noise vector, with different cluster following different Gaussian distributions. The left-hand figure shows the distributions of the measurement noise samples in the pixel plane. The measurement noise samples with the same color follow the same distribution. The right-hand figure presents the corresponding measurement noise distributions.

and $\hat{\boldsymbol{P}}_k(\lambda_k^{1:N})$ denotes posterior mean and covariance matrix related to $\lambda_k^{1:N}$, respectively. It is worth noting that the MNCM coefficients are independent of each other, thus their joint PDF can be written in the form of a product chain.

Since the variation tendency of the measurement noise is unknown and the MNCM coefficient λ_k^n is considered to be a slowly varying parameter, the prior model of the MNCM coefficients is established based on the posterior GIG distribution parameters at the time step $k-1$, scaled by the forgetting factor ρ. The joint prior PDF can be expressed as:

$$p(\boldsymbol{x}_k, \lambda_k^{1:N} | \boldsymbol{z}_{1:k-1}) = \mathcal{N}(\boldsymbol{x}; \bar{\boldsymbol{x}}_k, \bar{\boldsymbol{P}}_k) \prod_{n=1}^{N} GIG(\lambda_k^n; \bar{\alpha}_k^n, \bar{\beta}_k^n, \bar{\gamma}_k^n) \tag{4}$$

$$\begin{cases} \bar{\boldsymbol{x}}_k = \boldsymbol{\Phi}_k \hat{\boldsymbol{x}}_{k-1}, \bar{\boldsymbol{P}}_k = \boldsymbol{\Phi}_k \hat{\boldsymbol{P}}_{k-1} \boldsymbol{\Phi}_k^T + \boldsymbol{G}_k \boldsymbol{Q}_k \boldsymbol{G}_k^T \\ \bar{\alpha}_k^n = \rho \hat{\alpha}_{k-1}^n, \bar{\beta}_k^n = \rho \hat{\beta}_{k-1}^n, \bar{\gamma}_k^n = \rho \hat{\gamma}_{k-1}^n \end{cases} \tag{5}$$

C. EM-based Stepwise Adaptive Measurement Clustering

In the open outdoor scenarios, the number of measurement noise distributions remains unknown, and likewise, the specific distributions that the noise of each individual measurement adheres to are also not clear. Consequently, it becomes essential to conduct online clustering for the measurements on the basis of their noise distributions. Moreover, the number of clusters should be determined autonomously through the generation of new clusters. In this subsection, an EM algorithm-based stepwise adaptive measurement clustering method is introduced. In the first step, a chi-squared test will be conducted on the measurements for pre-clustering to reduce computational cost of iteration, while also enabling the autonomous generation of clusters. In the second step, the E-step of the EM algorithm is executed to calculate the probability of each measurement belonging to each candidate cluster. In the third step, the M-step of the EM algorithm is carried out to calculate the proportion of each cluster in the overall measurements.

The EM algorithm will cause significant computational costs since it requires iteratively calculating the probability that each measurement belongs to each cluster. To address the above issue, in the first step, a chi-squared test is utilized to exclude clusters that can not pass the test, and the calculations of probability that the measurement belongs to those clusters will be skipped, which indicates that the measurement does not belong to those clusters, thus the searching range will be reduced, as well as the computational costs. This step also serves to determine whether to generate a new cluster. The test statistic that m th measurement belongs to n th cluster is constructed as

$$O_k^{mn} = (\tilde{\boldsymbol{z}}_k^m)^T (\boldsymbol{P}_k^{zz(mn)})^{-1} \tilde{\boldsymbol{z}}_k^m \tag{6}$$

where the conditional distribution of m th measurement innovation $\tilde{\boldsymbol{z}}_k^m$ can be obtained as follows

$$\begin{cases} \tilde{\boldsymbol{z}}_k^m = \boldsymbol{z}_k^m - \boldsymbol{H}_k^m \bar{\boldsymbol{x}}_k, p(\tilde{\boldsymbol{z}}_k^m | \lambda_k^n) = \mathcal{N}(\tilde{\boldsymbol{z}}_k^m; \boldsymbol{0}, \boldsymbol{P}_k^{zz(mn)}) \\ \boldsymbol{P}_k^{zz(mn)} \approx \boldsymbol{H}_k^m \bar{\boldsymbol{P}}_k (\boldsymbol{H}_k^m)^T + \bar{\theta}_k^n \boldsymbol{I}_k \end{cases} \tag{7}$$

$\bar{\theta}_k^n$ denotes the prior mean of λ_k^n, which can be calculated by (5) and (22). By comparing the magnitude of the test statistic with the chi-square threshold, it can be determined whether the iterative computation of the probability that m th measurement belongs to n th cluster will be skipped. Moreover,

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to reduce the unnecessary computational burden caused by an excessive number of clusters, the maximum number of clusters is set to N_{max}. In the second step, the E-step of EM algorithm is implemented to calculate the probability that each measurement belongs to each candidate cluster. The hidden variable U_k^{nm} is introduced, which indicates whether the $m\ th$ measurement belongs to the $n\ th$ cluster, and $E(U_k^{nm}|\boldsymbol{z}_k^m, \lambda_k^n)$ reflects the probability that the $m\ th$ measurement belongs to the $n\ th$ cluster. Then there is

$$U_k^{nm} = \begin{cases} 1, m\ th\ measurement \in n\ th\ cluster \\ 0, else \end{cases} \tag{8}$$

The probability that the $m\ th$ measurement belongs to the $n\ th$ cluster can be calculated as

$$\begin{aligned} &E(U_k^{nm}|\boldsymbol{z}_k^m, \lambda_k^n) = p(U_k^{nm} = 1|\boldsymbol{z}_k^m, \lambda_k^n) \\ &= \frac{p(\boldsymbol{z}_k^m|U_k^{nm} = 1, \lambda_k^n)p(U_k^{nm} = 1|\lambda_k^n)}{\sum_{n=1}^{N^m} p(\boldsymbol{z}_k^m|U_k^{nm} = 1, \lambda_k^n)p(U_k^{nm} = 1|\lambda_k^n)} \\ &= \frac{\mathcal{N}(\boldsymbol{z}_k^m; \boldsymbol{H}_k^m \bar{\boldsymbol{x}}_k, \boldsymbol{P}_k^{zz(mn)})\pi_k^n}{\sum_{n=1}^{N^m} \mathcal{N}(\boldsymbol{z}_k^m; \boldsymbol{H}_k^m \bar{\boldsymbol{x}}_k, \boldsymbol{P}_k^{zz(mn)})\pi_k^n} \end{aligned} \tag{9}$$

where $\pi_k^n = p(U_k^{nm} = 1|\lambda_k^n)$, and it denotes the proportion of each cluster in the overall measurements. N^m represents the number of the candidate clusters that the $m\ th$ measurement likely belongs to. It should be emphasized that MSCKF employs a measurement delayed-update strategy, where $\boldsymbol{z}_k^m$ represents all historical measurement residuals during the observation period of the feature points, and it will facilitate correct clustering. In the third step, the M-step is implemented to solve for π_k^n as follows

$$\pi_k^n = \frac{\sum_{m=1}^{M} E(U_k^{nm}|\boldsymbol{z}_k^m, \lambda_k^n)}{M} \tag{10}$$

where M represents the total number of current measurements. Iterate (9) and (10) until convergence. After the iteration of EM algorithm , the $m\ th$ measurement will be assigned to the cluster with the highest probability, which can be expressed as:

$$\hat{n} = \underset{n=1,\ldots,N}{argmax}\, E(U_k^{nm}|\boldsymbol{z}_k^m, \lambda_k^n) \tag{11}$$

D. Analytical Update of Joint Posterior PDF

In this subsection, an analytical update method without fixed-point iteration is utilized to derive an approximate analytical solution of $p(\boldsymbol{x}_k, \lambda_k^{1:N}|\boldsymbol{z}_{1:k})$. Although the issue of the coupling between $\boldsymbol{x}_k$ and λ_k^n can be addressed utilizing fixed-point iteration within the VB method, the inherent correlation between the state vector and parameters may introduce significant approximation errors. Moreover, real-time performance will be impacted by fixed-point iteration. Therefore, it is necessary to infer their joint posterior PDF without neglecting the correlation between $\boldsymbol{x}_k$ and λ_k^n. The joint posterior PDF can be obtained as follows through (3) and Bayes' rule:

Algorithm 1 The implementation of analytical update.

Inputs: $\bar{\boldsymbol{x}}_k, \bar{\boldsymbol{P}}_k, \boldsymbol{z}_k, \boldsymbol{H}_k, \bar{\alpha}_k^{1:N}, \bar{\beta}_k^{1:N}, \bar{\gamma}_k^{1:N}, L, \epsilon$
begin
1. Set the initial iteration value of each cluster
for $n = 1$ **to** N **do**
  (1) $\alpha_k^{n(0)} = \bar{\alpha}_k^n, \beta_k^{n(0)} = \bar{\beta}_k^n, \gamma_k^{n(0)} = \bar{\alpha}_k^n$
  (2) $\hat{\theta}_k^{n(0)} = E(\lambda_k^n|\bar{\alpha}_k^n, \bar{\beta}_k^n, \bar{\gamma}_k^n)$
end for
2. Compute the posterior GIG parameters
for $n = 1$ **to** N **do**
  for $i = 1$ **to** L **do**
    (1) $\check{\boldsymbol{P}}_k^{zz(n)(i+1)} = \frac{1}{\hat{\theta}_k^{n(i)}} \boldsymbol{H}_k^n \bar{\boldsymbol{P}}_k (\boldsymbol{H}_k^n)^{\mathrm{T}} + \boldsymbol{I}_k^n$
    (2) $\delta_k^{n(i+1)} = \mathrm{tr}(\tilde{\boldsymbol{P}}_k^{zz} (\check{\boldsymbol{P}}_k^{zz(n)(i+1)})^{-1})$
    (3) $\hat{\alpha}_k^{n(i+1)} = \bar{\alpha}_k^n, \hat{\beta}_k^{n(i+1)} = \bar{\beta}_k^n + \delta_k^{n(i+1)}$
      $\hat{\gamma}_k^{n(i+1)} = \bar{\gamma}_k^n - 0.5 D_k^n$
    (4) $\theta_k^{n(i+1)} = E(\lambda_k^n|\hat{\alpha}_k^{n(i+1)}, \hat{\beta}_k^{n(i+1)}, \hat{\gamma}_k^{n(i+1)})$
    if $(\theta_k^{n(i+1)} - \theta_k^{n(i)})/\theta_k^{n(i+1)} < \epsilon$ **then**
      break
    end if
  end for
end for
3. Posterior update state
(1) $\hat{\boldsymbol{R}}_k = \boldsymbol{Diag}(\theta_k^{1(L_{max}^1)} \boldsymbol{I}_k^n, \ldots, \theta_k^{N(L_{max}^N)} \boldsymbol{I}_k^n)$
(2) $\check{\boldsymbol{P}}_k^{zz} = \boldsymbol{H}_k \bar{\boldsymbol{P}}_k (\boldsymbol{H}_k)^{\mathrm{T}} + \hat{\boldsymbol{R}}_k$
(3) $\tilde{\boldsymbol{z}}_k = \boldsymbol{z}_k - \boldsymbol{H}_k \bar{\boldsymbol{x}}_k, \boldsymbol{K}_k = \bar{\boldsymbol{P}}_k \boldsymbol{H}_k (\check{\boldsymbol{P}}_k^{zz})^{-1}$
(4) $\hat{\boldsymbol{x}}_k = \bar{\boldsymbol{x}}_k + \boldsymbol{K}_k \tilde{\boldsymbol{z}}_k, \hat{\boldsymbol{P}}_k = (\boldsymbol{I}_k - \boldsymbol{K}_k \boldsymbol{H}_k) \bar{\boldsymbol{P}}_k$
(5) $\hat{\alpha}_k^{1:N} = \hat{\alpha}_k^{1:N(L_{max})}, \hat{\beta}_k^{1:N} = \hat{\beta}_k^{1:N(L_{max})}$
  $\hat{\gamma}_k^{1:N} = \hat{\gamma}_k^{1:N(L_{max})}$
end
Outputs: $\hat{\boldsymbol{x}}_k$,$\hat{\boldsymbol{P}}_k$,$\hat{\alpha}_k^{1:N}$,$\hat{\beta}_k^{1:N}$,$\hat{\gamma}_k^{1:N}$

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$$\begin{aligned} &p(\boldsymbol{x}_k, \lambda_k^{1:N}|\boldsymbol{z}_{1:k}) \propto \prod_{n=1}^{N} \mathcal{N}(\boldsymbol{z}_k^n; \boldsymbol{H}_k^n \bar{\boldsymbol{x}}_k, \boldsymbol{P}_k^{zz(n)}(\lambda_k^n)) \times \\ &\mathcal{N}(\boldsymbol{x}_k; \hat{\boldsymbol{x}}_k(\lambda_k^{1:N}), \hat{\boldsymbol{P}}_k(\lambda_k^{1:N})) \prod_{n=1}^{N} GIG(\lambda_k^n; \bar{\alpha}_k^n, \bar{\beta}_k^n, \bar{\gamma}_k^n) \end{aligned} \tag{12}$$

$$\boldsymbol{P}_k^{zz(n)}(\lambda_k^n) = \boldsymbol{H}_k^n \bar{\boldsymbol{P}}_k (\boldsymbol{H}_k^n)^T + \boldsymbol{R}_k^n \tag{13}$$

The proof of (12) can be found in Appendix B. It is known that to make the joint posterior PDF at analytically solvable, $\boldsymbol{P}_k^{zz(n)}(\lambda_k^{1:N})$ must be scaled by λ_k^n. To this end, $\boldsymbol{P}_k^{zz(n)}(\lambda_k^{1:N})$ is approximated as

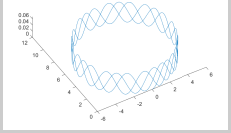
$$\boldsymbol{P}_k^{zz(n)}(\lambda_k^n) \approx \boldsymbol{H}_k^n \frac{\lambda_k^n}{\hat{\theta}_k^n} \bar{\boldsymbol{P}}_k (\boldsymbol{H}_k^n)^T + \lambda_k^n \boldsymbol{I}_k^n = \lambda_k^n \check{\boldsymbol{P}}_k^{zz(n)} \tag{14}$$

$\hat{\theta}_k^n$ denotes the posterior mean of λ_k^n, which can be calculated by utilizing (16) and (22). By employing (3), (14) and taking the logarithm of both sides of (12), it can be obtained with the terms related to λ_k^n as

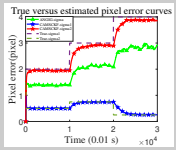
$$\begin{aligned} &\ln \mathcal{N}(\boldsymbol{x}_k; \hat{\boldsymbol{x}}_k(\lambda_k^{1:N}), \hat{\boldsymbol{P}}_k(\lambda_k^{1:N})) + (\hat{\gamma}_k^n - 1) \ln \lambda_k^n \\ &- 0.5(\hat{\alpha}_k^n \lambda_k^n + \frac{\hat{\beta}_k^n}{\lambda_k^n}) \approx \ln \mathcal{N}(\boldsymbol{x}_k; \hat{\boldsymbol{x}}_k(\lambda_k^{1:N}), \hat{\boldsymbol{P}}_k(\lambda_k^{1:N})) \\ &+ (\bar{\gamma}_k^n - 0.5 D_k^n - 1) \ln \lambda_k^n - 0.5(\bar{\alpha}_k^n \lambda_k^n + \frac{\bar{\beta}_k^n + \delta_k^n}{\lambda_k^n}) + c_{\lambda_k^n} \end{aligned} \tag{15}$$

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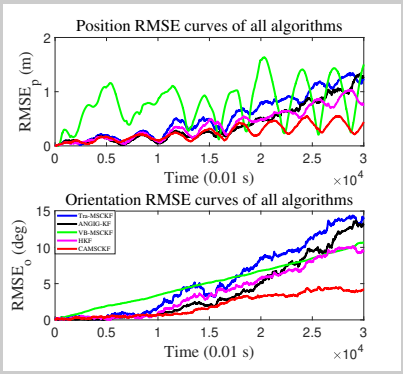
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(a) The actual motion trajectory of the robot



(b) Real and Estimated Pixel Error



(c) Position RMSE and Attitude RMSE curves of all methods in simulations

Fig. 4: Demonstration of the trajectory in simulation, pixel error curves and RMSE curves of all methods.

where $c_{\lambda_k^n}$ denotes a constant independent of λ_k^n, D_k^n denotes the dimension of $\boldsymbol{z}_{1:k}^n$ and δ_k^n is defined in (17). The posterior PDF of λ_k^n can be obtained through coefficients matching as follows

$$\begin{cases} p(\lambda_k^n|\boldsymbol{z}_{1:k}) = GIG(\lambda_k^n; \hat{\alpha_k^n}, \hat{\beta_k^n}, \hat{\gamma_k^n}) \\ \hat{\alpha_k^n} = \bar{\alpha_k^n}, \hat{\beta_k^n} = \bar{\beta_k^n} + \delta_k^n, \hat{\gamma_k^n} = \hat{\gamma_k^n} - 0.5D_k^n \end{cases} \tag{16}$$

$$\begin{cases} \delta_k^n = trace\{\tilde{\boldsymbol{P}}_k^{zz(n)}(\check{\boldsymbol{P}}_k^{zz(n)})^{-1}\} \\ \tilde{\boldsymbol{P}}_k^{zz(n)} = \tilde{\boldsymbol{z}}_k^n(\tilde{\boldsymbol{z}}_k^n)^T, \tilde{\boldsymbol{z}}_k^n = \boldsymbol{z}_k^n - \boldsymbol{H}_k^n\bar{\boldsymbol{x}}_k \end{cases} \tag{17}$$

By repeatedly iterating (16), the $\hat{\theta}_k^n$ can eventually converge to λ_k^n. Although an approximation error occurs when solving $p(\lambda_k^n|\boldsymbol{z}_{1:k})$ approximately, satisfactory accuracy can still be achieved, this will be proved in simulations and experiments in Section III and Section IV. After updating the MNCM coefficients, the approximate posterior PDF of the state vector can be obtained on this basis. The likelihood PDF is approximated as

$$\begin{cases} p(\boldsymbol{z}_k|\boldsymbol{x}_k, \lambda_k^{1:N}) \approx \mathcal{N}(\boldsymbol{z}_k; \boldsymbol{H}_k\boldsymbol{x}_k, \hat{\boldsymbol{R}}_k) \\ \hat{\boldsymbol{R}}_k = \boldsymbol{Diag}(\hat{\boldsymbol{R}}_k^1, ..., \hat{\boldsymbol{R}}_k^n, ..., \hat{\boldsymbol{R}}_k^N), \hat{\boldsymbol{R}}_k^n = \hat{\theta}_k^n\boldsymbol{I}_k^n \end{cases} \tag{18}$$

The conditional posterior PDF of $\boldsymbol{x}_k$ can be obtained as follows through (4), (18) and Bayes' rule

$$p(\boldsymbol{x}_k|\boldsymbol{z}_{1:k}, \lambda_k) = \mathcal{N}(\boldsymbol{x}_k; \hat{\boldsymbol{x}}_k, \hat{\boldsymbol{P}}_k) \tag{19}$$

$$\begin{cases} \hat{\boldsymbol{x}}_k = \bar{\boldsymbol{x}}_k + \boldsymbol{K}_k\tilde{\boldsymbol{z}}_k, \hat{\boldsymbol{P}}_k = (\boldsymbol{I}_k - \boldsymbol{K}_k\boldsymbol{H}_k)\bar{\boldsymbol{P}}_k \\ \boldsymbol{K}_k \approx \bar{\boldsymbol{P}}_k\boldsymbol{H}_k^T(\check{\boldsymbol{P}}_k^{zz})^{-1}, \check{\boldsymbol{P}}_k^{zz} = \boldsymbol{H}_k\bar{\boldsymbol{P}}_k\boldsymbol{H}_k^T + \hat{\boldsymbol{R}}_k \\ \tilde{\boldsymbol{z}}_k = \boldsymbol{z}_k - \boldsymbol{H}_k\bar{\boldsymbol{x}}_k \end{cases} \tag{20}$$

where $\boldsymbol{K}_k$ denotes the Kalman gain. The implementation for the joint posterior update is shown in Algorithm 1, where ϵ denotes the iteration termination lower bound, L_{max} denotes the maximum number of iterations.

III. SIMULATION COMPARISONS

In this section, the proposed method will be compared with other adaptive filter-based and robust filter-based VIO which can handle measurement noise with distinct and time-varying variances through simulations, including VB-MSCKF, HKF and ANGIG-KF. The parameter configurations are shown in Table I. In this scenario, a robot is set to execute a circular motion within the x-y plane, while moving up and down along the z-axis, as shown in Fig. 4(a). The feature points observed by the robot are randomly generated. The frequencies of the IMU and camera are 100 Hz and 10 Hz, respectively. The random walk coefficients of the gyroscope and accelerometer in the IMU, as well as the random walk coefficients of the constant biases, are set to be consistent with the Euroc dataset [29]. The measurement noise vector, i.e., pixel error, follows the multivariate Gaussian distributions with two different time-varying variances. The nominal variance of the pixel error does not match the actual one and the two variances even vary inconsistently over time. The actual standard deviations of the pixel errors are shown by dashed polylines *True-sigma1* and *True-sigma2* in Fig. 4(b). The total duration of the simulation is 30,000 steps (i.e., 300 seconds). 50 Monte Carlo simulations are performed for all methods to eliminate randomness. The position root mean square errors (PRMSE) and orientation root mean square errors (ORMSE) are shown in Fig. 4(c).

TABLE I: Parameter Configurations

VB-MSCKF	$\upsilon_0 = 10, a = 0.4, \boldsymbol{V}_0 = 0.1 * \boldsymbol{I}_6$
HKF	$\gamma = 1.345, M = 50, \epsilon = 1e^{-6}$
ANGIG-KF	$a = 1, b = 6.75, r = -0.5$
ACMSCKF	$\alpha_0 = 1, \beta_0 = 6.75, \gamma_0 = -0.5, N_{max} = 15$

As depicted in Fig. 4(c), the proposed ACMSCKF demonstrates the best accuracy and robustness among several methods compared, when the nominal MNCMs do not match the actual ones and several different variances vary inconsistently over time. It is believed that it benefits from the outstanding tracking ability of the proposed method for the MNCM, as shown in Fig. 4(b). The proposed ACMSCKF, can gradually converge to the true MNCM coefficients after a few iterations, achieving the best estimation performance. As for ANGIG-KF, it demonstrates a worse performance compared with ACMSCKF in estimating MNCM as shown in Fig. 4(b). It is attributed to its drawback regarding all elements of the measurement noise vector as the same distribution. Thus, the MNCM coefficients cannot be adjusted individually for the mismatch between the established MNCM model and the actual one, which can be intuitively presented in Fig. 4(b). Compared to the proposed method, the accuracy of Tra-MSCKF, HKF and VB-MSCKF is found to be rather limited, which struggle to deal with the scenario mentioned above, as shown in Fig. 4(c). It results from that Tra-MSCKF and HKF cannot track and adjust the MNCM adaptively, and VB-MSCKF is unstable due to its large approximation errors.

IV. EXPERIMENTS COMPARISONS

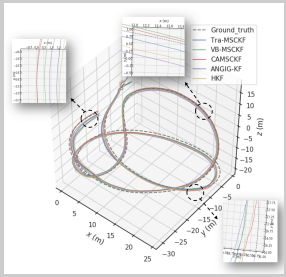
To further verify the superiority of the proposed ACMSCKF, experimental comparisons with the methods mentioned in Section III are conducted on the outdoor drone flight public dataset, which includes aggressive motion for VIO. All the outdoor sequences with publicly available

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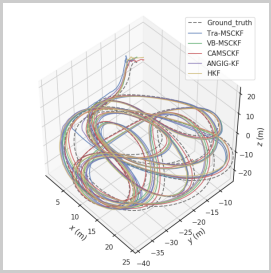
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TABLE II: ATEs and AOEs of All Tested UZH-FPV Outdoor Datasets

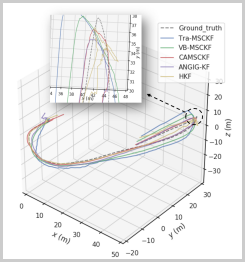
NO.Dataset	Performance metrics	Tra-MSCKF[6]	VB-MSCKF[24]	HKF[26]	ANGIG-KF[25]	ACMSCKF(Ours)
Outdoor_forward_1(easy)	AOE(deg)/ATE(m)/time(ms)	1.556/0.778/3.4	1.314/0.714/4.9	1.915/0.916/4.9	1.308/1.151/4.6	**0.886/0.487**/6.6
Outdoor_forward_3(medium)	AOE(deg)/ATE(m)/time(ms)	2.968/1.997/5.3	2.274/1.426/6.5	2.584/1.550/6.7	2.192/1.623/6.2	**1.684/1.154**/8.0
Outdoor_forward_5(hard)	AOE(deg)/ATE(m)/time(ms)	8.804/4.279/4.3	5.862/2.360/5.4	4.035/2.222/5.7	3.970/2.523/5.2	**2.571/1.896**/7.0
Outdoor_45	AOE(deg)/ATE(m)/time(ms)	3.224/0.603/4.9	2.434/1.333/5.5	3.350/0.700/5.3	2.840/0.531/5.7	**1.490/0.376**/7.0
Average	AOE(deg)/ATE(m)/time(ms)	4.138/1.914/4.5	2.971/1.458	2.980/1.347/5.6	2.578/1.457/5.7	**1.658/0.978**/7.2



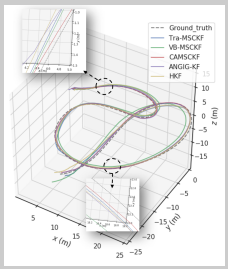
(a) Outdoor forward 1



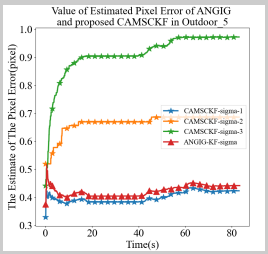
(b) Outdoor forward 3



(c) Outdoor forward 5



(d) Outdoor 45



(e) The curves of the estimated standard deviations of pixel errors in Outdoor_forward_5.

Fig. 5: All result in UZH-FPV outdoor sequences.

ground truths in UZH-FPV [30] datasets are selected for testing. Among them, sequences of three difficulty levels including simple, medium, and difficult are contained to comprehensively evaluate the proposed algorithm. All relevant equipment and parameters can be found in the introduction of the dataset. To ensure fairness, all methods adopt the same front-end (employing KLT for feature tracking) and utilize RANSAC for outlier rejection which is similar to OpenVINS [31]. Additionally, the number of the sliding window N is set to 11 and the maximum number of feature points involved in the measurement update at each time step is set to 40. The Absolute Trajectory Error (ATE), Absolute Orientation Error (AOE) and runtime of each method across these sequences will be evaluated as a basis for comparisons. Considering issues such as monocular scale drift, alignment is performed by utilizing the similarity transformation (Sim3). The nominal standard deviation of the visual measurement noise is set to 1.5 pixels. The parameter configurations of each method are listed in Table I and the results of all methods are shown in Table II and Fig. 5.

It can be seen from Table II that the proposed method achieves the best performance among all methods on the selected sequences, with an average ATE of 0.978m and an average AOE of 1.658°. It improves by 59.93% and 48.90% in the average AOE and ATE over all sequences compared with Tra-MSCKF, by 44.19% and 32.92% compared with VB-MSCKF, by 44.36% and 27.39% compared with HKF, and by 35.69% and 32.88% compared with ANGIG-KF. As depicted in Fig. 5(a)-(d), the proposed method holds the most consistent trajectories with the ground truths. It is believed that the proposed method benefits from the excellent tracking ability for MNCM, when the nominal variances of the measurement noise deviate from the actual ones and different variances vary inconsistently over time. As discussed earlier, Tra-MSCKF and HKF cannot track and adjust the MNCM adaptively and VB-MSCKF is unstable due to its large approximation errors. These crucial drawbacks are the principal reasons for their worse localization accuracy and robustness. As for ANGIG-KF, it is unable to track and adjust measurement noise variance that follows the multivariate Gaussian distribution with different time-varying variances individually, which leads to inferior performance. To further verify this point with the real data, the most challenging sequence $Outdoor_forward_5$ is selected for the MNCM coefficients estimation experiment and the results are demonstrated in Fig. 5(e). It is worth noting that only some of the representative MNCM estimation results by the proposed method are demonstrated. It can be seen that ANGIG-KF can only estimate the mean MNCM coefficient roughly, while ACMSCKF can cluster and estimate the MNCM coefficients efficiently according to the sample noise distributions. Benefiting from the more accurate MNCM coefficient estimates, the proposed method thus achieves better performance than that of ANGIG-KF. In addition, although the running time of the proposed method is longer, it can still fully meet the real-time requirements.

V. CONCLUSIONS

In this paper, an ACMSCKF for VIO is proposed, which can handle and track measurements with different noise variances continuously. Firstly, the joint distribution of the state vector and the MNCM coefficients for each cluster is modeled as an GMGIG distribution. Subsequently, an EM algorithm-based stepwise adaptive measurement clustering method is designed to achieve online cluster of measurements. Finally, an analytical update method for the joint posterior distribution without fixed-point iteration is carried out, it enables the tracking of the measurement noise distribution of each cluster and the automatic adjustment of the MNCM, achieving accurate and robust localization performance. Sufficient simulations and dataset experiments have

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been conducted, which verifies that the proposed method can significantly improve the localization accuracy and robustness of VIO with time-varying measurement noises.

VI. APPENDIX

A. GIG Distribution

The MNCM coefffcients λ of each measurement noise cluster is established obeying GIG distribution, the PDF and mean of λ can be presented as follows, respectively

$$GIG(\lambda;\alpha,\beta,\gamma)=\frac{(\frac{\alpha}{\beta})^{(\frac{\gamma}{2})}}{2K_r\sqrt{(\alpha\beta)}}\lambda^{\gamma-1}e^{\frac{-\alpha\lambda+\frac{\beta}{\lambda}}{2}} \tag{21}$$

$$E(\lambda)=\frac{\sqrt{\beta}K_{\gamma+1}(\sqrt{\alpha\beta})}{\sqrt{\alpha}K_{\gamma}(\sqrt{\alpha\beta})} \tag{22}$$

B. Proof of (13)

According to Bayes' rule and (3), it can be obtained as

$$\begin{aligned}&p(\boldsymbol{x}_k,\lambda_k^{1:N}|\boldsymbol{z}_{1:k})\propto p(\boldsymbol{z}_k|\boldsymbol{x}_k,\lambda_k^{1:N})p(\boldsymbol{x}_k,\lambda_k^{1:N}|\boldsymbol{z}_{1:k-1})\\&=p(\boldsymbol{z}_k|\boldsymbol{x}_k,\lambda_k^{1:N})p(\boldsymbol{x}_k|\boldsymbol{z}_{1:k-1},\lambda_k^{1:N})p(\lambda_k^{1:N}|\boldsymbol{z}_{1:k-1})\\&=p(\boldsymbol{z}_k|\boldsymbol{z}_{1:k-1},\lambda_k^{1:N})p(\boldsymbol{x}_k|\boldsymbol{z}_{1:k},\lambda_k^{1:N})p(\lambda_k^{1:N}|\boldsymbol{z}_{1:k-1})\end{aligned} \tag{23}$$

Utilizing Gaussian integral formula, by approximating $\boldsymbol{z}_k^n$ as mutually independent, it can be obtained as

$$\begin{aligned}&p(\boldsymbol{z}_k|\boldsymbol{z}_{1:k-1},\lambda_k^{1:N})\approx\prod_{i=1}^{N}p(\boldsymbol{z}_k^n|\boldsymbol{z}_{1:k-1}^n,\lambda_k^n)\\&=\prod_{i=1}^{N}\int\mathcal{N}(\boldsymbol{z}_k^n;\boldsymbol{H}_k^n\boldsymbol{x}_k,\lambda_k^n\boldsymbol{I}_k)\mathcal{N}(\boldsymbol{x}_k;\bar{\boldsymbol{x}}_k,\bar{\boldsymbol{P}}_k)dx_k\\&=\prod_{i=1}^{N}\mathcal{N}(\boldsymbol{z}_k^n;\boldsymbol{H}_k^n\bar{\boldsymbol{x}}_k,\boldsymbol{P}_k^{zz(n)})\end{aligned} \tag{24}$$

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